

Ian M. Nesbitt

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Williamstown, MA, USA

National Center for Ecological Analysis and Synthesis, U.C. Santa Barbara

DataONE Network Operations

Last modified: May 14, 2026

Education

2017–2022 **Masters Student in Earth and Climate Sciences**, *School of Earth and Climate Sciences, College of Natural Sciences, Forestry, and Agriculture, University of Maine, Orono, ME, USA*

GPA: 3.97

Thesis: Using diverse methods to determine the volume and stratigraphy of sediment in a formerly glaciated lake in central Maine, USA

Advisor: [Seth W. Campbell](#), *Co-advisor:* [Sean M.C. Smith](#)

Associated Projects: [readgssi](#), a Python tool to read and process radar data; [SeidarT](#), a seismic and radar survey modeling toolbox

2009–2013 **Bachelor of Arts in Geosciences with Honors**, *Williams College, Williamstown, MA, USA*

Thesis: A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range

Advisor: [David P. Dethier](#)

2006–2009 **Degree with Honors**, *Holderness School, Holderness, NH, USA*

Awards and Scholarships

2021 Golden Key (invited)

2020 US Antarctica Service Medal

2013 Sigma Xi induction

2012 Williams College Class of 1960 Scholar in Geosciences

Volunteer Experience

Since 2025 Co-Chair, Earth Science Information Partners ([ESIP](#)) [schema.org](#) Cluster with [Colin Smith](#)

Since 2022 Nordic Ski Coach, Mt. Greylock Regional School

2018–2023 Co-President, Williams College Class of 2013

2017–2018 Co-Director of Reunion Planning, Williams College Class of 2013

Work Experience

- Since 2022 **Research Software Engineer**, *National Center for Ecological Analysis and Synthesis (NCEAS), U.C. Santa Barbara, Santa Barbara, CA, USA (remote)*
Duties: Python and Java development of open-source computing tools to interpret, quality-check, archive, and serve large quantities of research science data. Working on **DataONE Federation** metadata interoperability and data processing tools. DataONE Network Operations duties include monitoring and troubleshooting of **DataONE** member nodes and coordinating with the community to ensure the health of the network.
Supervisor: **Matt Jones**
- Since 2022 **Assistant Head Coach**, *Berkshire Nordic Ski Club, Williamstown, MA, USA*
Duties: Group and one-on-one summer Nordic ski training, technique instruction, goal setting, customized training plans. Planning and coordination with other coaches, athletes, and parents.
Head Coach: Hilary Greene
- 2022 **Geospatial Engineering Consultant**, *Muddy Water Initiative, Boston, MA, USA*
Duties: Geospatial database management, design of SQL joins and relates, QA/QC and digitization of field data, presentation-quality map design, client and stakeholder interaction for **Caroline Reeves**.
- 2021 **Geophysicist**, *International Thwaites Glacier Collaboration, Project GHC, Orono, ME, USA*
Duties: Processed ground-penetrating radar (GPR) survey data from the Hudson Mountains, West Antarctica, in collaboration with **John Woodward**, **Joanne S. Johnson**, **Brent Goehring**, and **Seth W. Campbell**.
- 2021 **Research Software Developer**, *University of Maine, Orono, ME, USA*
Duties: Tested and debugged code for geophysical modeling software **SeidarT**, developed a multi-platform install script for **Christopher C. Gerbi** and **Steven P. Bernsen**.
Projects: <https://github.com/UMainedynamics/seidart>
- 2019–2020 **Lead Geophysicist**, *Allan Hills Antarctic Expedition I-165-M, Princeton University - NSF Award Number 1744993, Allan Hills Field Camp, East Antarctica*
Duties: Collected and processed 100 km of ground-penetrating radar (GPR) survey data from a blue ice area on the eastern side of the Trans-Antarctic Mountains. Deep field survival, data collection, processing, use of snow machines in harsh conditions.
- 2018–2020 **Lead Scientist**, *Raspberry Shake, S.A., Boquete, Chiriqui, Panama*
Duties: Technical support, software development, QA/QC of seismic instruments, development and QA/QC of educational materials.
Projects: <https://github.com/raspishake/rsudp>, <https://edu.raspberrysshake.org>
- 2014–2017 **Geophysical Scientist**, *e4sciences LLC, Newtown, CT, USA*
Duties: Led field team for many types of geophysical survey including Mobile LiDAR, ground-penetrating radar, high-precision GPS, multibeam echosounding, sub-bottom seismic, sidescan sonar. Field interpretation, and on-the-ground operations decision-making including boat handling.
Projects: Created survey-to-delivery workflow for 3D LiDAR end-user deliverables.

- 2013–2014 **Assistant Nordic Ski Coach**, *St. Michael's College*, Colchester, VT, USA
Duties: Created and supervised individualized training plans for athletes, technique instruction, race day logistics including ski preparation.

Teaching

- 2019 **Ground-Penetrating Radar Theory and Applications to Geological Mapping – SC2**, *Northeastern Section Meeting, Geological Society of America*, Portland, ME, USA
Seth W. Campbell and Ian M. Nesbitt (filling in for Steven A. Arcone) taught short course at NEGSA annual meeting, covering radar application, data collection, processing, and interpretation.
- 2018 **Global Environmental Change – ERS 201**, *University of Maine*, Orono, ME, USA
Karl J. Kreutz and Ian M. Nesbitt instructed students in accepted and debated climate theory from an earth systems perspective.
- 2017 **Earth Systems – ERS 200**, *University of Maine*, Orono, ME, USA
Aaron E. Putnam, Sean M.C. Smith, Peter O. Koons, and Ian M. Nesbitt instructed students in earth systems theory.
- 2012–2013 **GIS and Remote Sensing – GEOS 214**, *Williams College*, Williamstown, MA, USA
David P. Dethier and Ian M. Nesbitt instructed students in traditional and emerging geomorphologic theory, methods, and practice.
- 2012 **Geomorphology – GEOS 201**, *Williams College*, Williamstown, MA, USA
David P. Dethier and Ian M. Nesbitt instructed students in geographical information systems (GIS) theory, technology and software, and advised in the application of these methods via end-of-term projects.

Publications

Peer-reviewed scientific articles

- in review **The A-KBS Arctic Knowledge Based System: Science Gateway Integration for Exascale Arctic Data Processing and Geospatial Feature Prediction**, *ESS Open Archive*
Meisam Shayeghmoradi, Andrew Wilcox, Sheridan Parker, Aaron Bergstrom, Ian M. Nesbitt, Naima Kaabouch, and Timothy Pasch
preprint: [10.22541/essoar.177292734.41397214/v1](https://doi.org/10.22541/essoar.177292734.41397214/v1)
- 2026 **Broadly stable atmospheric CO₂ and CH₄ levels over the past 3 million years**, *Nature*
J. Marks-Peterson, Sarah Shackleton, John Higgins et al.
doi: [10.1038/s41586-025-10032-y](https://doi.org/10.1038/s41586-025-10032-y)

- 2026 **The arctic knowledge-based system: Science gateway integration for petascale arctic data processing and geospatial feature prediction**, *Applied Computing and Geosciences*
Andrew Wilcox, Meisam Shayeghmoradi, Stephen Miller, [Ian M. Nesbitt](#), Saisri Pogalla, Aymane Ahajjam, Walker McKee, Sheridan Parker, Matthew Johnson, [Aaron Bergstrom](#) et al.
doi: [10.1016/j.acags.2026.100322](#)
- 2025 **Assessing the suitability of sites near Pine Island Glacier for subglacial bedrock drilling aimed at detecting Holocene retreat–readvance**, *The Cryosphere*
[Joanne S. Johnson](#), [John Woodward](#), [Ian M. Nesbitt](#), [Kate Winter](#), [Seth W. Campbell](#), [Kier Nichols](#), [Ryan Venturelli](#), [Scott Braddock](#), [Brent Goehring](#), [Brenda L. Hall](#) et al.
doi: [10.5194/tc-19-303-2025](#)
- 2025 **Miocene and Pliocene ice and air from the Allan Hills blue ice area, East Antarctica**, *Proceedings of the National Academy of Sciences*
[Sarah Shackleton](#), [Valens Hishamunda](#), L. Davidge, [Ed Brook](#), J. Marks-Peterson, [Austin Carter](#), S. Aarons, [Andrei Kurbatov](#), [Douglas Introne](#), Y. Yan, [Ian M. Nesbitt](#) et al.
doi: [10.1073/pnas.2502681122](#)
- in review **The sediment delivery continuum from deglaciation to the modern watershed based on lake sedimentary deposits in the Northeastern USA**, *Quaternary Research*
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Sean M.C. Smith](#), [Bess G. Koffman](#), [Steven A. Arcone](#), and [Kristin M. Schild](#)
preprint: [10.13140/RG.2.2.11567.05281](#)
- 2021 **rsudp: A Python package for real-time seismic monitoring with Raspberry Shake instruments**, *Journal of Open Source Software*
[Ian M. Nesbitt](#), Richard I. Boaz, and Justin Long
doi: [10.21105/joss.02565](#)
- 2020 **Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures**, *Science*
Thomas Lecocq, Stephen P Hicks, Koen Van Noten, Kasper van Wijk, Paula Koelemeijer, Raphael SM De Plaen, Frédéric Massin, Gregor Hillers, Robert E Anthony, Maria-Theresia Apoloner, ... [Ian M. Nesbitt](#) et al.
doi: [10.1126/science.abd2438](#)
- 2016 **Tracing subarctic Pacific water masses with benthic foraminiferal stable isotopes during the LGM and late Pleistocene**, *Deep-Sea Research Part II: Topical Studies in Oceanography*
[Mea S. Cook](#), A. Christina Ravelo, Alan C. Mix, [Ian M. Nesbitt](#), and Nari V. Miller
doi: [10.1016/j.dsr2.2016.02.006](#)

Conference proceedings, presentations, and datasets

- 2025 **Validation and interoperability of schema.org as a science metadata standard: a DataONE perspective**, *Earth Science Information Partners (ESIP) Annual Meeting*, Seattle, WA, USA, Invited talk
[Ian M. Nesbitt](#)
[Speaker page](#)
- 2022 **Post-glacial sediment delivery continuum to an impounded valley reach in central Maine: a multi-disciplinary approach**, *Geological Society of America Abstracts with Programs*, Invited talk
[Ian M. Nesbitt](#), [Sean M.C. Smith](#), [Seth W. Campbell](#), [Bess G. Koffman](#), [Steven A. Arcone](#), and [Kristin M. Schild](#)
doi: [10.1130/abs/2022NE-374798](https://doi.org/10.1130/abs/2022NE-374798)
- 2021 **Unprecedented present deglaciation of Pine Island Glacier Inferred from Ice Penetrating Radar Studies of Localised Ice Domes in the Hudson Mountains**, *International Thwaites Glacier Collaboration Conference*, Poster
[John Woodward](#), [Joanne S. Johnson](#), [Seth W. Campbell](#), [Ian M. Nesbitt](#), [Brenda L. Hall](#), [Scott Braddock](#), [Meghan Spoth](#), [Brent Goehring](#), [Ryan Venturelli](#), [Dylan H. Rood](#), [Kier Nichols](#), [Johnathan Adams](#), and [Greg Balco](#)
- 2019 **A decision-making framework for sedimentation analyses in dammed river corridor impoundments**, *Geological Society of America Abstracts with Programs 51*, Portland, ME, Talk
[Ian M. Nesbitt](#), [Sean M.C. Smith](#), [Bess G. Koffman](#), [Seth W. Campbell](#), and [Steven A. Arcone](#)
doi: [10.1130/abs/2019NE-328587](https://doi.org/10.1130/abs/2019NE-328587)
- 2018 **Sedimentary architecture and accumulation rates of multiple lakes in New England, USA**, *AGU Fall Meeting Abstracts 2018*, Washington, D.C., Poster NS41B-0830
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), [Sean M.C. Smith](#), and [Bess G. Koffman](#)
- 2018 **Sedimentary architecture of Maine lakes derived with ground-penetrating radar**, *Maine Water and Sustainability Conference*, Augusta, ME, Poster
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), and [Sean M.C. Smith](#)
- 2018 **Holocene sediment volume determined by ground-penetrating radar and sidescan sonar in Maine, USA**, *Geological Society of America, Northeastern Section - 53rd Annual Meeting*, Talk
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), and [Sean M.C. Smith](#)
doi: [10.1130/abs/2018NE-311370](https://doi.org/10.1130/abs/2018NE-311370)

- 2017 **Using ground-penetrating radar and sidescan sonar to compare lake bottom geology in New England**, *AGU Fall Meeting Abstracts 2017*, New Orleans, LA, [Talk PP44B-01](#)
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), and [Sean M.C. Smith](#)
- 2017 **New England lake bottom geology and sedimentation rates derived from ground-penetrating radar**, *GSA Annual Meeting*, Seattle, WA
[Seth W. Campbell](#), [Ian M. Nesbitt](#), [Steven A. Arcone](#), [Sean M.C. Smith](#)
doi: [10.1130/abs/2017AM-306235](#)
- 2015 **Geophysical imaging of dams and levees in the Northeast US**, *Society of Exploration Geophysicists Abstracts With Programs*, New Orleans, LA
Daniel A. Rosales, William F. Murphy IV, Matthew B. Art, [Ian M. Nesbitt](#), Salvatore Triano, David C. Herron, Kurt Schollmeyer, James Trotta, W. Bruce Ward, Lisa Stewart, and William F. Murphy III
- 2014 **Tracing Bering Sea circulation with benthic foraminiferal stable isotopes during the Pleistocene**, *AGU Fall Meeting Abstracts 2014*, PP23D-08
[Mea S. Cook](#), [A. Christina Ravelo](#), [Alan C. Mix](#), [Ian M. Nesbitt](#), and [Nari V. Miller](#)
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Thesis, Department of Geosciences, Williams College*, Williamstown, MA
[Ian M. Nesbitt](#)
Advisor: [David P. Dethier](#)
Available via [Williams College Libraries](#)
doi: [10.13140/RG.2.2.33197.20966](#)
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Geological Society of America, Northeastern Section - 48th Annual Meeting*, Bretton Woods, NH
[Ian M. Nesbitt](#) and [David P. Dethier](#)
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Twenty-sixth annual Keck Research Symposium in Geology, proceedings 2013*, Pomona, CA
[Ian M. Nesbitt](#) and [Robert Varga](#)

Open-source software

- 2025 **MetacatUI 2.36.0**, *Arctic Data Center*
[Matt Jones](#), [Chris Jones](#), [Lauren Walker](#), [Robyn Thiessen-Bock](#), [Ben Leinfelder](#), [Peter Slaughter](#), [Bryce Mecum](#), [Rushiraj Nenuji](#), [Hasham Elbashandy](#), [Val Hendrix](#) et al.
<https://github.com/NCEAS/MetacatUI>
doi: [10.18739/a2hx15s97](#)

- 2025 **MetacatUI 2.35.0**, *Arctic Data Center*
 Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
<https://github.com/NCEAS/MetacatUI>
 doi: 10.18739/a2q52ff87
- 2024 **HashStore — Python**, *Arctic Data Center*
 DouMing Mok, Matthew Brooke, Jing Tao, Jeanette Clark, Matt Jones, and Ian M. Nesbitt
<https://github.com/DataONEorg/hashstore>
 doi: 10.18739/a2zg6g87q
- 2024 **MetacatUI 2.34.0**, *Arctic Data Center*
 Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
<https://github.com/NCEAS/MetacatUI>
 doi: 10.18739/a2w669b2r
- 2021 **SeidarT: Seismic and radar survey modeling toolbox**
 Steven P. Bernsen, Christopher C. Gerbi, Ian M. Nesbitt, Ann Hill, Senthil Vel, Knut Christianson, Seth W. Campbell, and Ben Hills
<https://github.com/UMainedynamics/SeidarT>
 doi: 10.5281/zenodo.5498194
- 2020 **rsudp: Continuous visual display, sudden motion monitoring, and historical replay of Raspberry Shake data**
 Ian M. Nesbitt, Richard I. Boaz, and Justin Long
<https://github.com/raspishake/rsudp>
 doi: 10.21105/joss.02565
- 2019 **gpx2dzg: a tool to convert GPX files to GSSI's proprietary DZG format, and plot comparisons of marks in GPX and DZT/DZX files**, *Zenodo*
 Developer and maintainer
<https://github.com/iannesbitt/gpx2dzg>
 doi: 10.5281/zenodo.3260948
- 2019 **readgssi: an open-source tool to read and plot GSSI ground-penetrating radar data**, *Zenodo*
 Developer and maintainer
<https://github.com/iannesbitt/readgssi>
 doi: 10.5281/zenodo.1439119

Technical Skills

Languages	Python, R, Matlab, Node.js, Java, Ruby/Rails
Markup	Django Templates, HTML, CSS, L ^A T _E X, Markdown, Overleaf
Platform Tools	Docker, Kubernetes, Selfhosted Linux servers, Raspberry Pi, Heroku, AWS
Developer Tools	GNU/Linux, Unix Terminal/bash/zsh, GNU Make, git, GitHub, Jupyter Notebooks, Visual Studio Code, Apple Developer Tools

Bug Tracking	Request Tracker, MantisBT, GitHub
APIs	DataONE API , FDSN , USGS Earthquakes/EarthExplorer, Telegram, Google Maps, Google Workspace
Other Tools	QGIS, ArcGIS, Inkscape, GIMP, Illustrator, Photoshop, ReflexW, RADAN
Communication	Slack, Discord, Mattermost
Office Suites	Microsoft Office, LibreOffice, Google Docs/Sheets/Slides

Spoken Languages

English	Native
Spanish	Beginner

Certifications

Since 2022	SafeSport Trained
2013, 2018	Wilderness First-Aid
2016	CPR, First-Aid, and Defibrillator
2019	Antarctic Deep Field Survival
2020	T3 Alliance certified instructor